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Research Profile

MSc Cognitive Systems student at Ulm University with research interests in Human-Computer Interaction, Human-AI Interaction, cognitive load, adaptive interfaces, and immersive simulation. My work combines user-centered design, interactive system prototyping, experimental thinking, and technical implementation to study how people perceive, trust, and interact with intelligent systems.

Research Interests

Human-Computer Interaction; Human-AI Interaction; Cognitive Load and Mental Workload; Adaptive Interfaces; AR/VR and Immersive Simulation; Trust in Intelligent Systems; Physiological and Behavioral Measures in HCI; Responsible AI and Human-Centered AI Governance; Usable Security and Privacy; UX Research and Usability Evaluation.

Education

MSc Cognitive Systems, Ulm University, Germany — Oct 2023 – Present

- Focus: Human-Computer Interaction, cognitive systems, interaction design, neurotechnology, and human-centered intelligent systems.
- Relevant coursework: Foundations and Concepts of Cognitive Systems Modeling (1.0), Neurotechnology: Brain-Machine Interfacing (1.0), Cognition in Complex Situations (1.3), User-Centered Design for Interactive Systems I & II (1.3), and Design Thinking for Interactive Systems (1.3).
- Research direction: cognitive load, trust, adaptive interfaces, immersive simulation, and human interaction with intelligent systems.

BSc (Hons) Software Engineering, University of Plymouth, United Kingdom — Mar 2018 – Jul 2021

- Awarded Second Class Honours, Upper Division.
- Foundation in software engineering, web development, databases, distributed systems, HCI, and project-based development.

Chinese Language Foundation Course, Nanjing Normal University, China — Sep 2016 – Jul 2017

Research Experience

Master's Thesis Work, Ulm University — April 2026 – Present

- Investigating low-cost VR vehicle motion simulators for studying perception, user experience, and motion-related cues in simulated vehicle environments.
- Research themes: VR simulation, human-vehicle interaction, cognitive load, motion perception, user experience, and low-cost experimental platforms.

Selected Research Projects

Trust Calibration and Adaptive Visualization in Automated Driving — 2025 – Present

- Developed a Unity-based experimental simulation to study trust, mental demand, and user perception in automated driving scenarios.
- Integrated questionnaire-based user feedback with adaptive visualization and Bayesian optimization concepts.
- Research themes: human-automation interaction, trust calibration, cognitive load, adaptive interfaces, and simulation-based HCI.
- Ongoing online study: Visualization of Automated Vehicle Functionalities.

Assessing Cognitive Load in Immersive Environments 2025

- Investigated cognitive load assessment in immersive and VR-based environments using human-centered evaluation methods.
- Explored subjective workload measures and physiological or behavioral indicators for evaluating interactive systems.
- Research themes: cognitive load, immersive HCI, user studies, workload measurement, and experimental design.

Unity Prototype for Trust Calibration in Highly Automated Driving 2024 – 2025

- Developed a Unity-based research prototype exploring trust calibration in highly automated driving through reliability scenarios and visual feedback concepts.
- Designed a planned 3×3 experimental structure combining visualization approach (optimized, static, no visualization) with system reliability level (low, medium, high).
- Implemented scenario concepts involving pedestrian crossings, false positives, false negatives, occlusion, covered areas, trajectory cues, and segmentation-based highlighting.
- Prepared technical foundations for studying trust, cognitive load, and visual appeal in automated-driving interfaces.
- Project status: prototype and study-design implementation completed; participant study not conducted in this version.

Binaural Audio and Motion-Based Experimental Automation 2024 – 2025

- Supported the automation of audio playback, recording, and motion-tracking workflows for controlled cognitive and auditory experiments.
- Worked with multichannel audio recordings, experimental timing, structured data saving, and spectrogram generation.
- Research themes: auditory perception, signal processing, and experimental automation.

Academic Writing and Presentations

- Seminar paper: *Critical Analysis and Reflection of Concept Representation in Language Models*.
- Research project report: *Assessing Cognitive Load in Immersive Environments*.
- Case study: *Challenges in Partially Automated Driving and a Proposed Human-Centered Solution*.

Research and Technical Skills

Research Methods: Experimental design, human-subjects experiment execution, questionnaire-based evaluation, cognitive load assessment, usability evaluation, literature review.

HCI & Design: Human-centered design, UX research, interaction design, accessibility, wireframing, prototyping, A/B testing.

Programming & Prototyping: Python, C#, Unity, JavaScript, TypeScript, HTML, CSS, PHP, SQL.

Data & Analysis: Experimental data pipeline development, data collection workflows, data preprocessing, data cleaning, structured data storage, CSV/Excel-based workflows, signal processing basics, statistical analysis basics, and data visualization.

Tools: Unity, GitHub, Figma, Adobe XD, VS Code, TYPO3, WordPress, OpenAI Codex, Docker basic, Azure DevOps basic.

University and Technical Experience

Student Assistant, Ulm University May 2026 – Present

Executive Support Unit: Quality Development, Reporting and Revision

- Develop and maintain TYPO3-based web pages following Ulm University's digital communication and content standards.
- Contribute to Python-based development and automation tasks supporting reporting, data preparation, and workflow improvement.

- Support data analysis and administrative processes by organizing, validating, and preparing information for reporting and quality development.

Student Assistant – Marketing, Ulm University — Oct 2025 – Present
Central Student Advisory Service

- Design, develop, and maintain TYPO3-based web pages for student-facing information and university communication.
- Improve content structure, readability, and user journeys for prospective and current students.
- Collaborate with advisory and communication teams to update event information, verify links, and support bilingual content presentation.

Student Assistant – Web Development, University Hospital Ulm — May 2024 – Present
MOCA Lab

- Maintain and update research-related websites using WordPress and TYPO3.
- Support responsive design, performance optimization, and content layout improvements.
- Collaborate with research staff to align web content with project goals.

Student Assistant – Web Development, Ulm University — Jan 2025 – Dec 2025
i3R Center

- Designed, developed, and maintained TYPO3-based web pages for research and institutional communication.
- Implemented frontend components and optimized templates for accessibility, usability, and performance.

Professional Experience

Front-End Engineer, WSO2 — Nov 2021 – Nov 2023

- Developed responsive and accessible web interfaces, landing pages, and marketing platforms using HTML, CSS, JavaScript, jQuery, and Bootstrap.
- Designed and prototyped user-centered web and email experiences using Figma and Adobe XD.
- Conducted usability-oriented improvements and A/B testing using tools such as Hotjar, GA4, and Pardot.
- Contributed to large-scale website and portal redesigns, including information architecture, UI design, and implementation.
- Collaborated across design, engineering, and marketing teams in an agile development environment.

Marketing Design Intern, WSO2 — May 2021 – Sep 2021

- Built reusable design assets and icon libraries aligned with brand goals.
- Developed mobile-friendly email templates with improved user flow and visual consistency.

Awards and Recognition

- Top 5% Global – WSO2 Performance Review, 2022 H1.
- Top 20% Global – WSO2 Performance Review, 2022 H2.
- National Rank 39 in Arts Stream – GCE Advanced Level Examination, Sri Lanka, 2015.

Languages

English – Fluent | Sinhala – Native

References

Available upon request.